

Numpy 2 Hacker Rank

1. Task

Your task is to print an array of size X with its main diagonal elements as 's and 's everywhere else.

Note

In order to get alignment correct, please insert the line

```
np.set_printoptions(legacy='1.13')
```

below the numpy import.

Input Format

A single line containing the space separated values of and .

N denotes the rows.

M denotes the columns.

Output Format

Print the desired N X Marray.

Sample Input

```
3 3
```

Sample Output

```
[[ 1.  0.  0.]
```

```
 [ 0.  1.  0.]
```

```
 [ 0.  0.  1.]]
```

Change Theme Language Pypy 3

```
1 # Enter your code here. Read input from STDIN. Print output to STDOUT
2 import numpy as np
3
4 n,m=map(int,input().strip().split())
5 np.set_printoptions(legacy='1.13')
6 print(np.eye(n,m))
```



You have earned 20.00 points!

You are now 125 points away from the gold level for your python badge.

31%

275/400

Congratulations

You solved this challenge. Would you like to challenge your friends?



Next Challenge

Test case 0

Compiler Message

Success

Test case 1

Test case 2

Input (stdin)

Download

1 3 3

Expected Output

Download

1 [[1. 0. 0.]

2 [0. 1. 0.]

3 [0. 0. 1.]]

2.

Task

You are given two integer arrays, A and B of dimensions $N \times M$.

Your task is to perform the following operations:

1. Add ($A + B$)
2. Subtract ($A - B$)
3. Multiply ($A * B$)
4. Integer Division (A / B)
5. Mod ($A \% B$)
6. Power ($A ** B$)

Note

There is a method `numpy.floor_divide()` that works like `numpy.divide()` except it performs a floor division.

Input Format

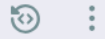
The first line contains two space separated integers, N and M .

The next N lines contains M space separated integers of array A .

The following N lines contains M space separated integers of array B .

Output Format

Print the result of each operation in the given order under **Task**.



```
1 # Enter your code here. Read input from STDIN. Print output to STDOUT
2 import numpy as np
3 n,m =map(int,input().strip().split())
4 arr =[]
5 arr1 =[]
6 for _ in range(n):
7     arr.append(list(map(int,input().strip().split(' '))))
8 for _ in range(n):
9     arr1.append(list(map(int,input().strip().split(' '))))
10
11 a = np.array(arr)
12 b = np.array(arr1)
13
14 print(a+b)
15 print(a-b)
16 print(a*b)
17 print(a//b)
18 print(a%b)
19 print(a**b)
20
```



You have earned 20.00 points!

You are now 105 points away from the gold level for your python badge.

42%

295/400

Congratulations

You solved this challenge. Would you like to challenge your friends?



Next Challenge

✓ Test case 0

Compiler Message

Success

✓ Test case 1

✓ Test case 2

Input (stdin)

```
1 1 4
2 1 2 3 4
3 5 6 7 8
```

[Download](#)

Expected Output

```
1 [[ 6  8 10 12]]
2 [[-4 -4 -4 -4]]
3 [[ 5 12 21 32]]
```

[Download](#)

3.

Task

You are given a 1-D array, A . Your task is to print the *floor*, *ceil* and *rint* of all the elements of A .

Note

In order to get the correct output format, add the line

`numpy.set_printoptions(legacy='1.13')` below the numpy import.

Input Format

A single line of input containing the space separated elements of array A .

Output Format

On the first line, print the *floor* of A .

On the second line, print the *ceil* of A .

On the third line, print the *rint* of A .

Sample Input

```
1.1 2.2 3.3 4.4 5.5 6.6 7.7 8.8 9.9
```

Sample Output

```
[ 1.  2.  3.  4.  5.  6.  7.  8.  9.]  
[ 2.  3.  4.  5.  6.  7.  8.  9. 10.]  
[ 1.  2.  3.  4.  6.  7.  8.  9. 10.]
```

```

1  # Enter your code here. Read input from STDIN. Print output to STDOUT
2  import numpy as np
3
4  np.set_printoptions(legacy='1.13')
5  arr = np.array(list(map(float, input().strip().split()))))
6
7  print(np.floor(arr))
8  print(np.ceil(arr))
9  print(np rint(arr))

```



You have earned 20.00 points!

You are now 85 points away from the gold level for your python badge.

53%

315/400

Congratulations

You solved this challenge. Would you like to challenge your friends?



Next Challenge

Test case 0

Compiler Message

Success

Test case 1

Test case 2

Input (stdin)

Download

```
1 1.1 2.2 3.3 4.4 5.5 6.6 7.7 8.8 9.9
```

Expected Output

Download

```

1  [ 1.  2.  3.  4.  5.  6.  7.  8.  9.]
2  [ 2.  3.  4.  5.  6.  7.  8.  9. 10.]
3  [ 1.  2.  3.  4.  6.  7.  8.  9. 10.]

```